

github link: https://github.com/Ameer-Hamja/Sports-Questions_vtu30434

Initial Arrangement

Position : 1 2 3 4 5

Board : 2 3 1 5 4

Students : 1 2 3 4 5

After Beat 1

Initially:

Students : 1 2 3 4 5

Movement:

Student 1 → position 2

Student 2 → position 3

Student 3 → position 1

Student 4 → position 5

Student 5 → position 4

New arrangement:

Students : 3 1 2 5 4

After Beat 2

Current arrangement:

Students : 3 1 2 5 4

Again follow the board:

Student 3 → position 2

Student 1 → position 3

Student 2 → position 1

Student 5 → position 5

Student 4 → position 4

New arrangement:

Students : 2 3 1 4 5

After Beat 3

Students : 1 2 3 5 4

After Beat 4

Students : 3 1 2 4 5

After Beat 5

Students : 2 3 1 5 4

After Beat 6

Students : 1 2 3 4 5

Original arrangement is restored.

LCM METHOD

Cycle 1

$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$

Length = 3

Cycle 2

$4 \rightarrow 5 \rightarrow 4$

Length = 2

The first cycle repeats every 3 beats, and the second repeats every 2 beats.

The smallest beat where both cycles finish together is:

$\text{LCM}(3, 2) = 6$

Hence, the answer is 6 drum beats.